Lab 8

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Task 1

- `apps/bookmodule/urls.py` — added route:

```
path('lab8/task1', views.lab8_task1, name="books.lab8_task1"),
```

- `apps/bookmodule/views.py` — imported `Q` and added view at the bottom:

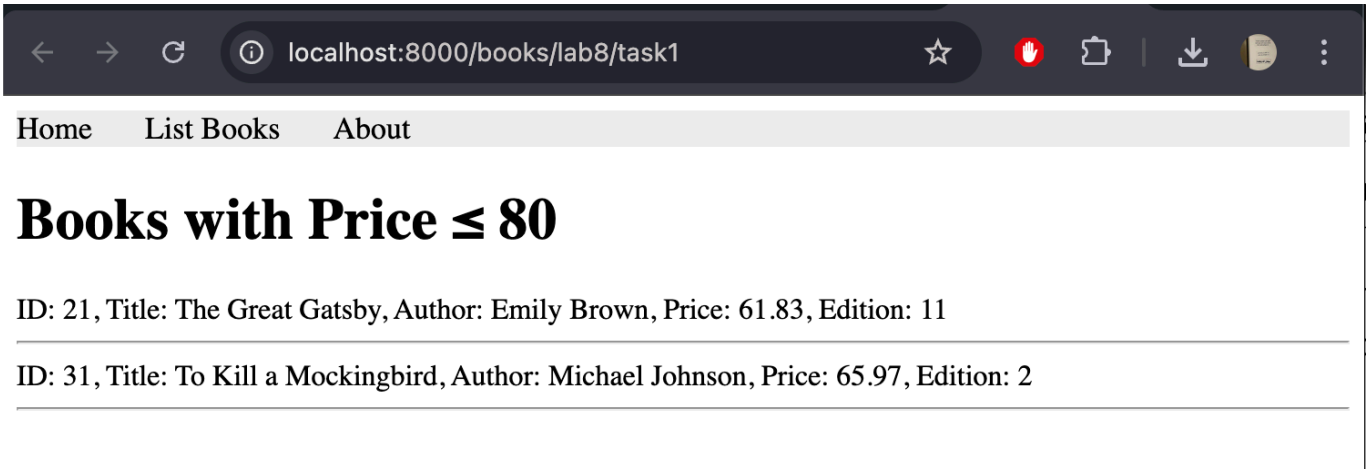
```
from django.db.models import Q

def lab8_task1(request):
    mybooks = Book.objects.filter(Q(price__lte=80))
    return render(request, 'bookmodule/lab8_task1.html', {'books':
mybooks})
```

- `apps/templates/bookmodule/lab8_task1.html` — new template that displays the filtered books, with an `{% empty %}` fallback if none are found.

```
{% extends "layouts/base.html" %}
{% block title %} Lab8 Task1 {% endblock title %}
{% block content %}
<h1>Books with Price &le; 80</h1>
{% for book in books %}
ID: {{ book.id }}, Title: {{ book.title }}, Author: {{ book.author }},
Price: {{ book.price }}, Edition: {{ book.edition }}
<hr>
{% empty %}
<p>No books found with price less than or equal to 80.</p>
{% endfor %}
{% endblock content %}
```

Output at `http://localhost:8000/books/lab8/task1`:



Task 2

- `apps/bookmodule/urls.py` — added route:

```
path('lab8/task2', views.lab8_task2, name="books.lab8_task2"),
```

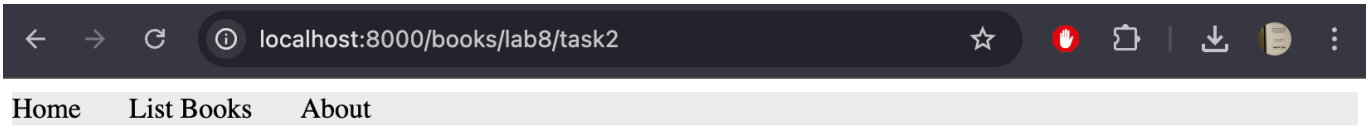
- `apps/bookmodule/views.py` — added view at the bottom:

```
def lab8_task2(request):
    mybooks = Book.objects.filter(Q(edition__gt=3) &
(Q(title__icontains='qu') | Q(author__icontains='qu')))
    return render(request, 'bookmodule/lab8_task2.html', {'books':
mybooks})
```

- `apps/templates/bookmodule/lab8_task2.html` — new template displaying the filtered books, with an `{% empty %}` fallback if none are found.

```
{% extends "layouts/base.html" %}
{% block title %} Lab8 Task2 {% endblock title %}
{% block content %}
<h1>Books with Edition &gt; 3 and Title or Author Contains 'qu'</h1>
{% for book in books %}
  ID: {{ book.id }}, Title: {{ book.title }}, Author: {{ book.author
}}, Price: {{ book.price }}, Edition: {{ book.edition }}
  <hr>
{% empty %}
  <p>No books found matching the criteria.</p>
{% endfor %}
{% endblock content %}
```

Output at `http://localhost:8000/books/lab8/task2`:



Task 3

- `apps/bookmodule/urls.py` — added route:

```
path('lab8/task3', views.lab8_task3, name="books.lab8_task3"),
```

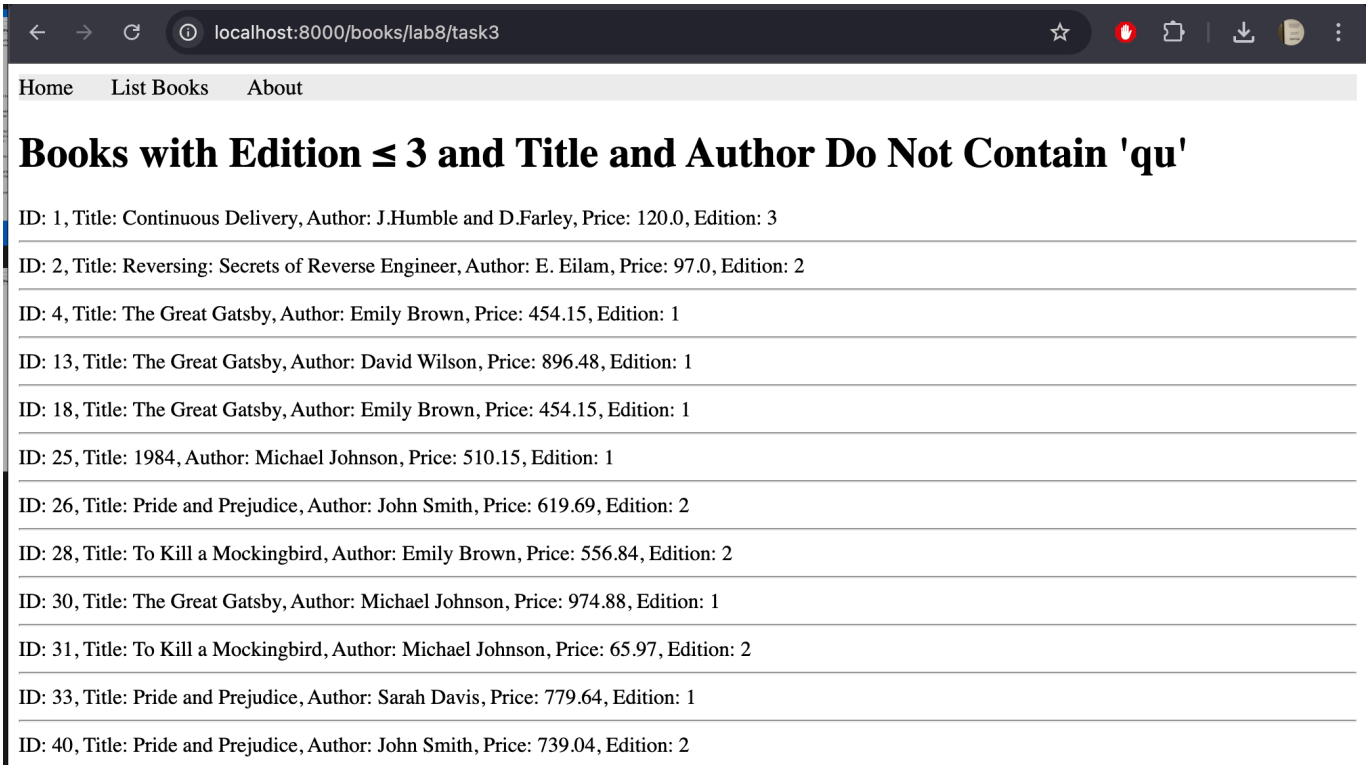
- `apps/bookmodule/views.py` — added view at the bottom:

```
def lab8_task3(request):
    mybooks = Book.objects.filter(~Q(edition__gt=3) & ~
(Q(title__icontains='qu') | Q(author__icontains='qu')))
    return render(request, 'bookmodule/lab8_task3.html', {'books':
mybooks})
```

- `apps/templates/bookmodule/lab8_task3.html` — new template displaying the filtered books, with an `{% empty %}` fallback if none are found.

```
{% extends "layouts/base.html" %}
{% block title %} Lab8 Task3 {% endblock title %}
{% block content %}
<h1>Books with Edition &le; 3 and Title and Author Do Not Contain
'qu'</h1>
{% for book in books %}
  ID: {{ book.id }}, Title: {{ book.title }}, Author: {{ book.author
}}, Price: {{ book.price }}, Edition: {{ book.edition }}
  <hr>
{% empty %}
  <p>No books found matching the criteria.</p>
{% endfor %}
{% endblock content %}
```

Output at `http://localhost:8000/books/lab8/task3`:



Task 4

- `apps/bookmodule/urls.py` — added route:

```
path('lab8/task4', views.lab8_task4, name="books.lab8_task4"),
```

- `apps/bookmodule/views.py` — added view at the bottom:

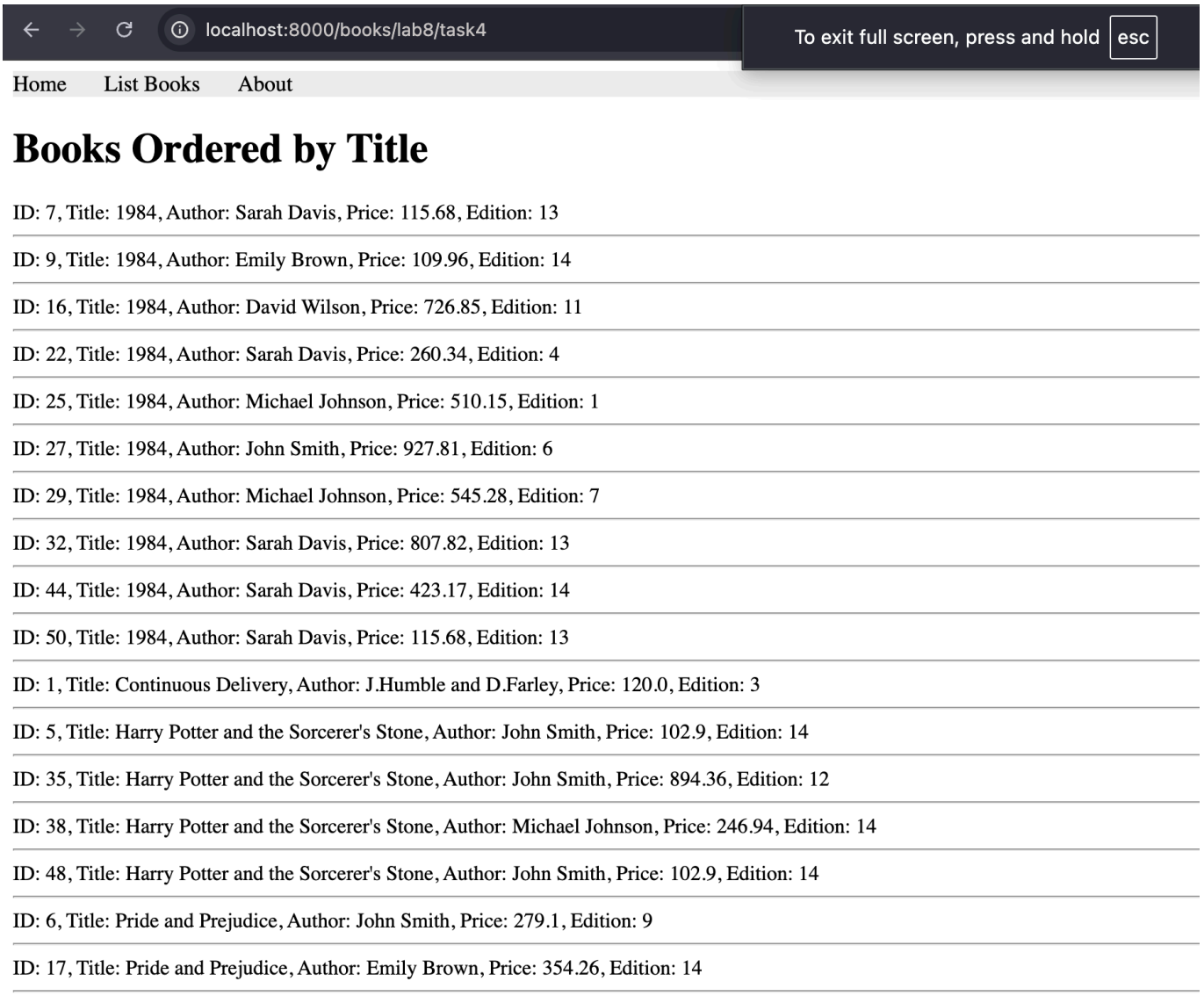
```
def lab8_task4(request):
    mybooks = Book.objects.all().order_by('title')
    return render(request, 'bookmodule/lab8_task4.html', {'books':
mybooks})
```

- `apps/templates/bookmodule/lab8_task4.html` — new template displaying books sorted by title, with an `{% empty %}` fallback if none are found.

```
{% extends "layouts/base.html" %}
{% block title %} Lab8 Task4 {% endblock title %}
{% block content %}
<h1>Books Ordered by Title</h1>
{% for book in books %}
  ID: {{ book.id }}, Title: {{ book.title }}, Author: {{ book.author
}}, Price: {{ book.price }}, Edition: {{ book.edition }}
  <hr>
{% empty %}
  <p>No books found.</p>
```

```
{% endfor %}
{% endblock content %}
```

Output at `http://localhost:8000/books/lab8/task4`:



Task 5

Created URL `/books/lab8/task5` that displays aggregated book statistics (count, total, average, max, and min price) using Django aggregation functions.

Files changed:

- `apps/bookmodule/urls.py` — added route:

```
path('lab8/task5', views.lab8_task5, name="books.lab8_task5"),
```

- `apps/bookmodule/views.py` — imported aggregation functions and added view at the bottom:

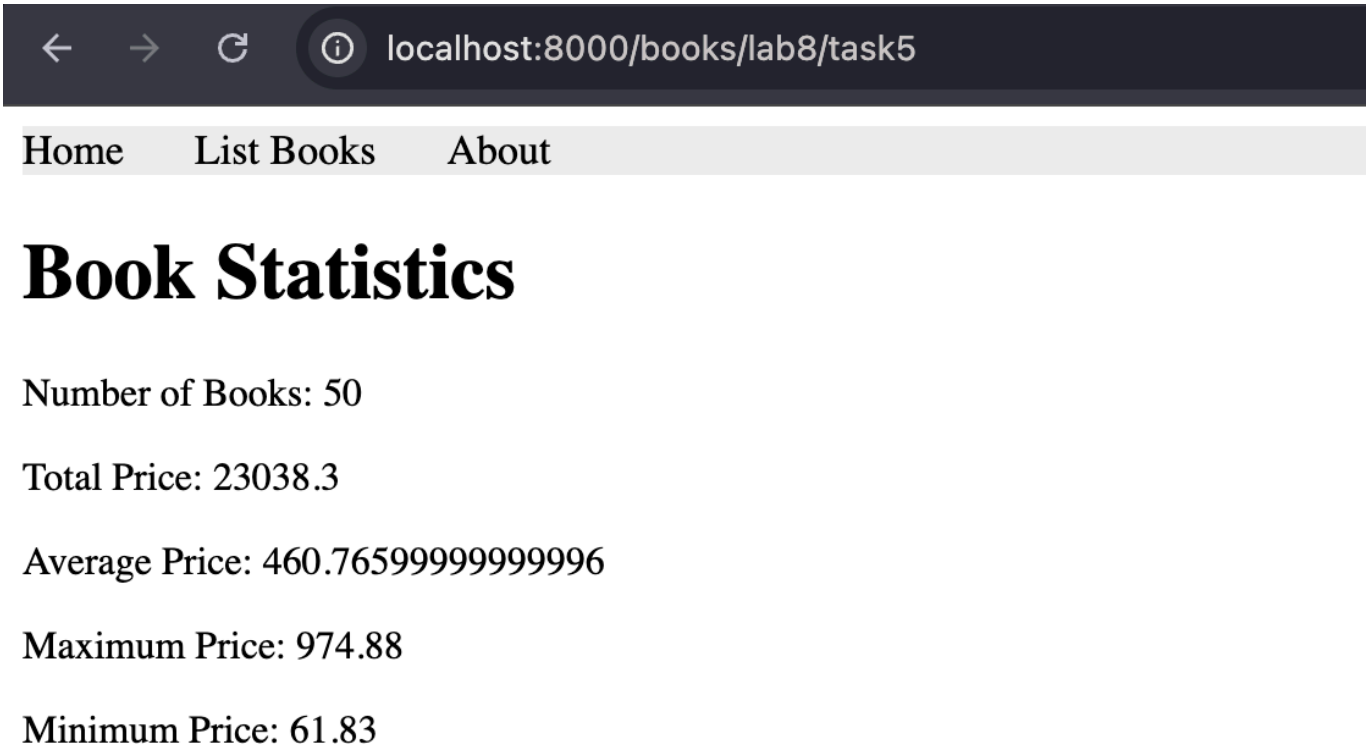
```
from django.db.models import Q, Count, Sum, Avg, Max, Min

def lab8_task5(request):
    stats = Book.objects.aggregate(
        count=Count('id'),
        total_price=Sum('price'),
        avg_price=Avg('price'),
        max_price=Max('price'),
        min_price=Min('price'),
    )
    return render(request, 'bookmodule/lab8_task5.html', {'stats':
stats})
```

- `apps/templates/bookmodule/lab8_task5.html` — new template displaying the aggregated statistics.

```
{% extends "layouts/base.html" %}
{% block title %} Lab8 Task5 {% endblock title %}
{% block content %}
<h1>Book Statistics</h1>
<p>Number of Books: {{ stats.count }}</p>
<p>Total Price: {{ stats.total_price }}</p>
<p>Average Price: {{ stats.avg_price }}</p>
<p>Maximum Price: {{ stats.max_price }}</p>
<p>Minimum Price: {{ stats.min_price }}</p>
{% endblock content %}
```

Output at `http://localhost:8000/books/lab8/task5`:



Task 6

- `apps/usermodule/models.py` — added two models with a ForeignKey relationship:

  ```
  from django.db import models

  class Address(models.Model):
      city = models.CharField(max_length=100)

  class Student(models.Model):
      name = models.CharField(max_length=100)
      age = models.IntegerField()
      address = models.ForeignKey(Address, on_delete=models.SET_NULL,
  null=True)
  ```

- `apps/usermodule/migrations/0001_initial.py` — auto-generated migration that creates both tables in the database.

Migrations applied with:

```
python manage.py makemigrations
python manage.py migrate
```

Task 7

- `apps/bookmodule/urls.py` — added route:

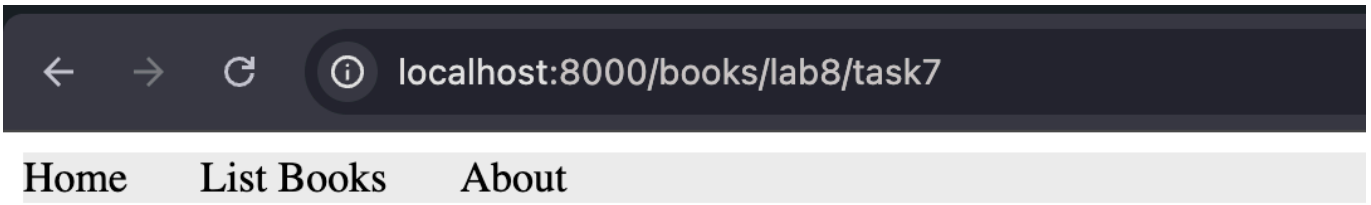
```
path('lab8/task7', views.lab8_task7, name="books.lab8_task7"),
```

- `apps/bookmodule/views.py` — imported `Address` from `usermodule` and added view at the bottom:

```
from apps.usermodule.models import Address

def lab8_task7(request):
    cities = Address.objects.annotate(student_count=Count('student'))
    return render(request, 'bookmodule/lab8_task7.html', {'cities':
cities})
```

- `apps/templates/bookmodule/lab8_task7.html` — new template displaying each city and its student count, with an `{% empty %}` fallback if none are found.

```
{% extends "layouts/base.html" %}
{% block title %} Lab8 Task7 {% endblock title %}
{% block content %}
<h1>Number of Students per City</h1>
{% for address in cities %}
  City: {{ address.city }}, Number of Students: {{
address.student_count }}
  <hr>
{% empty %}
  <p>No cities found.</p>
{% endfor %}
{% endblock content %}
```

Output at `http://localhost:8000/books/lab8/task7`: